

Additional validation and CHL weight prediction

1 Additional validation

This study examines width, depth and optimizer dependence on two representative power families. The planned depth/width study comprises 78 seeded neural fits. This report contains 78 distinct completed fits. Uniform networks have one, three or six hidden ReLU layers; the original Net1 and Net3 are included as references. The wider and deeper architectures are compared on the same data splits and training budget within each study.

All test metrics are evaluated after predictions have been returned to the signed physical weight w :

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \frac{|\hat{w}_i - w_i|}{|w_i|}, \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{w}_i - w_i)^2}. \quad (2)$$

The CHL datasets have nonzero weights. Thus their MAPE requires no omission of zero targets and is not a percentage error of a logarithm. When a logarithmic target is fitted, its transformation is inverted before these metrics are computed, including the known sign and family-dependent conversion from power to weight. Training and validation losses remain in the fitted target space; their numerical values should not be compared across different target transformations. Recorded input transformations: **L2**, **LogAbsL2**. Recorded target transformations: **LogAbsPower**, **StandardizedWeight**. Recorded training budgets: 10000 rounds. For **LogAbsL2**, coefficient magnitudes are logged and the resulting vector is normalized to unit Euclidean norm; exactly zero coefficients are mapped to zero while retaining their positions. For **L2**, raw coefficient vectors are normalized to unit Euclidean norm. Standardized weights use only the training-set mean and sample standard deviation; the same transformation is applied to validation and test data. For **LogAbsPower**, the fitted target is $\log |s|$, where s is the power; a predicted value z is decoded as $\hat{w} = \text{sgn}(s) K_0 e^z$, with the known weight K_0 of the parent form. The final checkpoint and the checkpoint minimizing validation loss are reported separately. Test errors play no part in checkpoint selection. Error bars are sample standard deviations over training seeds on fixed data partitions, not confidence intervals over alternative partitions or modular-form families.

Positive CHL powers, $N = 3$. Among completed three-seed configurations, the lowest mean validation-selected test MAPE is obtained by Net3 with ADAM: $0.400 \pm 0.0987\%$.

Among completed three-seed configurations, the lowest mean validation-selected test RMSE is obtained by Net3 with ADAM: 12.8 ± 2.35 .

The same configuration minimizes both reported mean test metrics within this grid.

At width 16 with RMSProp, changing from one to three hidden layers changes mean test MAPE from 5.06% to 0.851%, and mean test RMSE from 148 to 34.0.

At width 128 with RMSProp, changing from one to three hidden layers changes mean test MAPE from 0.699% to 0.765%, and mean test RMSE from 31.2 to 33.5.

For 1×128 , Adam and RMSProp respectively give mean test MAPE 3.73% and 0.699%, and mean test RMSE 113 and 31.2. This comparison concerns the recorded optimizer settings, not independently tuned optimizer families.

For Net3, Adam and RMSProp respectively give mean test MAPE 0.400% and 0.998%, and mean test RMSE 12.8 and 40.3. This comparison concerns the recorded optimizer settings, not independently tuned optimizer families.

Comparisons of different depths also change parameter count; they do not isolate depth at fixed capacity. Test rankings are descriptive and are not used to choose checkpoints or tune hyperparameters.

Negative eta powers. Among completed three-seed configurations, the lowest mean validation-selected test MAPE is obtained by 6×128 with ADAM: $0.0509 \pm 0.00801\%$.

Among completed three-seed configurations, the lowest mean validation-selected test RMSE is obtained by 6×128 with ADAM: 0.0345 ± 0.00147 .

The same configuration minimizes both reported mean test metrics within this grid.

At width 16 with ADAM, changing from one to three hidden layers changes mean test MAPE from 1.29% to 0.316%, and mean test RMSE from 0.725 to 0.260.

At width 128 with ADAM, changing from one to three hidden layers changes mean test MAPE from 0.153% to 0.0510%, and mean test RMSE from 0.0698 to 0.0392.

For 1×128 , Adam and RMSProp respectively give mean test MAPE 0.153% and 0.185%, and mean test RMSE 0.0698 and 0.160. This comparison concerns the recorded optimizer settings, not independently tuned optimizer families.

For Net1, Adam and RMSProp respectively give mean test MAPE 42.6% and 0.147%, and mean test RMSE 20.1 and 0.123. This comparison concerns the recorded optimizer settings, not independently tuned optimizer families.

Comparisons of different depths also change parameter count; they do not isolate depth at fixed capacity. Test rankings are descriptive and are not used to choose checkpoints or tune hyperparameters.

Training budget and checkpoint choice. Both checkpoint conventions are tabulated. The largest ratio of final to validation-selected mean test RMSE among complete configurations is 10.1, for Positive CHL powers, $N = 3$, Net3, ADAM (129 versus 12.8). Validation selection does not guarantee lower error on every held-out test set.

Capacity and interpolation. At the final checkpoint, 2 of 78 completed fits have training NRMSE at most 10^{-3} , with NRMSE normalized by the training-set standard deviation of the physical weight. This threshold is a numerical convention for near interpolation. A double-descent claim would additionally require a reproducible test-error peak near a resolved interpolation transition and a subsequent decline. Neither having more parameters than examples nor an isolated non-monotone error curve establishes that claim.

1.1 Validation-selected checkpoints

$D \times h$ / model	Optimizer	P	n	Train loss	Val. loss	MAPE (%)	RMSE
1×1	RMSProp	33	3	0.620	0.826	102 ± 74.8	1406 ± 875
1×4	RMSProp	129	3	0.321	0.425	55.8 ± 77.5	839 ± 934
1×8	RMSProp	257	3	0.00498	0.00444	5.54 ± 2.27	165 ± 66.6
1×16	RMSProp	513	3	0.00436	0.00404	5.06 ± 2.92	148 ± 83.6
1×24	RMSProp	769	3	7.13×10^{-4}	8.33×10^{-4}	2.07 ± 0.681	62.4 ± 21.2
3×16	RMSProp	1057	3	9.44×10^{-5}	1.17×10^{-4}	0.851 ± 0.0594	34.0 ± 0.871
6×16	RMSProp	1873	3	1.99×10^{-4}	1.75×10^{-4}	1.19 ± 0.0621	49.2 ± 9.92
1×128	ADAM	4097	3	0.00286	0.00277	3.73 ± 2.77	113 ± 89.8
1×128	RMSProp	4097	3	1.03×10^{-4}	1.63×10^{-4}	0.699 ± 0.134	31.2 ± 8.00
3×128	RMSProp	37121	3	7.16×10^{-5}	1.14×10^{-4}	0.765 ± 0.148	33.5 ± 3.83
6×128	RMSProp	86657	3	5.27×10^{-5}	7.21×10^{-5}	0.618 ± 0.106	29.4 ± 3.49
Net3	ADAM	133745	3	9.10×10^{-6}	1.22×10^{-5}	0.400 ± 0.0987	12.8 ± 2.35
Net3	RMSProp	133745	3	2.68×10^{-4}	1.26×10^{-4}	0.998 ± 0.473	40.3 ± 9.28

Table 1: Positive CHL powers, $N = 3$. Validation-selected checkpoint. Uniform architectures are labelled by hidden-layer count D and width h ; Net1 and Net3 retain their original tapered shapes. P is the number of trainable parameters and n the number of completed training seeds. Training and validation loss are in the fitted target space. Test MAPE is on signed weights and is reported as a percentage; test RMSE is in weight units. Test entries are mean $\pm$ sample standard deviation. A single completed seed has no estimated standard deviation. Training budget: 10000 rounds.

$D \times h$ / model	Optimizer	P	n	Train loss	Val. loss	MAPE (%)	RMSE
1×1	ADAM	33	3	0.359	0.453	55.5 ± 62.6	27.5 ± 28.1
1×4	ADAM	129	3	0.00334	0.00354	6.12 ± 0.643	3.51 ± 0.606
1×8	ADAM	257	3	8.25×10^{-4}	9.55×10^{-4}	3.01 ± 1.97	1.66 ± 0.990
1×16	ADAM	513	3	1.52×10^{-4}	1.87×10^{-4}	1.29 ± 0.292	0.725 ± 0.236
1×24	ADAM	769	3	5.23×10^{-5}	5.77×10^{-5}	0.817 ± 0.295	0.444 ± 0.177
3×16	ADAM	1057	3	1.72×10^{-5}	1.66×10^{-5}	0.316 ± 0.0164	0.260 ± 0.0485
6×16	ADAM	1873	3	5.35×10^{-6}	7.34×10^{-6}	0.162 ± 0.0189	0.142 ± 0.0216
1×128	ADAM	4097	3	8.80×10^{-7}	1.79×10^{-6}	0.153 ± 0.0235	0.0698 ± 0.0103
1×128	RMSProp	4097	3	5.55×10^{-6}	5.10×10^{-6}	0.185 ± 0.0306	0.160 ± 0.0445
3×128	ADAM	37121	3	3.91×10^{-7}	5.25×10^{-7}	0.0510 ± 0.0160	0.0392 ± 0.00611
Net1	ADAM	51473	3	0.332	0.423	42.6 ± 73.7	20.1 ± 34.5
Net1	RMSProp	51473	3	3.67×10^{-6}	9.10×10^{-6}	0.147 ± 0.0418	0.123 ± 0.0320
6×128	ADAM	86657	3	2.68×10^{-7}	4.48×10^{-7}	0.0509 ± 0.00801	0.0345 ± 0.00147

Table 2: Negative eta powers. Validation-selected checkpoint. Uniform architectures are labelled by hidden-layer count D and width h ; Net1 and Net3 retain their original tapered shapes. P is the number of trainable parameters and n the number of completed training seeds. Training and validation loss are in the fitted target space. Test MAPE is on signed weights and is reported as a percentage; test RMSE is in weight units. Test entries are mean $\pm$ sample standard deviation. A single completed seed has no estimated standard deviation. Training budget: 10000 rounds.

1.2 Final checkpoints

$D \times h$ / model	Optimizer	P	n	Train loss	Val. loss	MAPE (%)	RMSE
1×1	RMSProp	33	3	0.620	0.828	99.0 ± 72.4	1432 ± 889
1×4	RMSProp	129	3	0.321	0.427	54.5 ± 75.3	898 ± 907
1×8	RMSProp	257	3	0.00634	0.00556	6.08 ± 2.79	168 ± 70.1
1×16	RMSProp	513	3	0.00472	0.00465	5.20 ± 2.90	179 ± 99.4
1×24	RMSProp	769	3	0.00167	0.00178	3.50 ± 1.23	127 ± 50.0
3×16	RMSProp	1057	3	0.00111	0.00136	2.73 ± 1.73	129 ± 72.9
6×16	RMSProp	1873	3	0.00125	0.00171	3.27 ± 1.23	130 ± 58.7
1×128	ADAM	4097	3	0.00298	0.00306	3.72 ± 2.82	111 ± 76.3
1×128	RMSProp	4097	3	0.00109	0.00120	2.92 ± 1.07	125 ± 50.2
3×128	RMSProp	37121	3	6.67×10^{-4}	0.00106	2.24 ± 1.46	97.3 ± 60.1
6×128	RMSProp	86657	3	0.00108	0.00157	2.49 ± 1.80	112 ± 75.6
Net3	ADAM	133745	3	0.00120	0.00162	3.06 ± 1.95	129 ± 92.7
Net3	RMSProp	133745	3	0.00325	0.00389	4.93 ± 2.72	203 ± 116

Table 3: Positive CHL powers, $N = 3$. Final checkpoint. Uniform architectures are labelled by hidden-layer count D and width h ; Net1 and Net3 retain their original tapered shapes. P is the number of trainable parameters and n the number of completed training seeds. Training and validation loss are in the fitted target space. Test MAPE is on signed weights and is reported as a percentage; test RMSE is in weight units. Test entries are mean $\pm$ sample standard deviation. A single completed seed has no estimated standard deviation. Training budget: 10000 rounds.

$D \times h$ / model	Optimizer	P	n	Train loss	Val. loss	MAPE (%)	RMSE
1×1	ADAM	33	3	0.359	0.453	55.5 ± 62.6	27.5 ± 28.1
1×4	ADAM	129	3	0.00333	0.00357	6.17 ± 0.709	3.49 ± 0.604
1×8	ADAM	257	3	8.12×10^{-4}	9.73×10^{-4}	3.06 ± 2.01	1.67 ± 0.960
1×16	ADAM	513	3	1.36×10^{-4}	2.03×10^{-4}	1.32 ± 0.375	0.713 ± 0.251
1×24	ADAM	769	3	5.89×10^{-5}	7.08×10^{-5}	0.839 ± 0.296	0.446 ± 0.209
3×16	ADAM	1057	3	2.67×10^{-5}	3.40×10^{-5}	0.351 ± 0.0703	0.330 ± 0.0691
6×16	ADAM	1873	3	3.03×10^{-4}	3.50×10^{-4}	0.739 ± 0.512	0.748 ± 0.853
1×128	ADAM	4097	3	2.46×10^{-6}	4.38×10^{-6}	0.165 ± 0.0236	0.0968 ± 0.0201
1×128	RMSProp	4097	3	6.57×10^{-5}	7.28×10^{-5}	0.610 ± 0.622	0.409 ± 0.319
3×128	ADAM	37121	3	2.16×10^{-5}	2.54×10^{-5}	0.302 ± 0.390	0.182 ± 0.208
Net1	ADAM	51473	3	0.332	0.423	42.8 ± 73.6	20.1 ± 34.5
Net1	RMSProp	51473	3	2.81×10^{-4}	3.22×10^{-4}	0.989 ± 0.543	0.890 ± 0.553
6×128	ADAM	86657	3	1.23×10^{-6}	2.44×10^{-6}	0.0741 ± 0.0257	0.0655 ± 0.0152

Table 4: Negative eta powers. Final checkpoint. Uniform architectures are labelled by hidden-layer count D and width h ; Net1 and Net3 retain their original tapered shapes. P is the number of trainable parameters and n the number of completed training seeds. Training and validation loss are in the fitted target space. Test MAPE is on signed weights and is reported as a percentage; test RMSE is in weight units. Test entries are mean $\pm$ sample standard deviation. A single completed seed has no estimated standard deviation. Training budget: 10000 rounds.

Scope. These experiments measure prediction within restricted power families. They do not establish extrapolation to new modular-form families or the necessity of neural networks. For full eta coefficient vectors, the weight is recovered exactly from $-a_1/(2a_0)$. For CHL eta products at $N > 1$, the corresponding predictor is $-a_1/a_0$; at $N = 1$ it is $-a_1/(2a_0)$. These identities are elementary baselines, not learned results, and apply when the required coefficients are retained.

2 CHL tables

All test metrics are evaluated after predictions have been returned to the signed physical weight w :

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \frac{|\hat{w}_i - w_i|}{|w_i|}, \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{w}_i - w_i)^2}. \quad (4)$$

The CHL datasets have nonzero weights. Thus their MAPE requires no omission of zero targets and is not a percentage error of a logarithm. When a logarithmic target is fitted, its transformation is inverted before these metrics are computed, including the known sign and family-dependent conversion from power to weight. Training and validation losses remain in the fitted target space; their numerical values should not be compared across different target transformations. Recorded input transformations: **LogAbsL2**. Recorded target transformations: **LogAbsPower**. Recorded training budgets: 10000 rounds. For **LogAbsL2**, coefficient magnitudes are logged and the resulting vector is normalized to unit Euclidean norm; exactly zero coefficients are mapped to zero while retaining their positions. For **LogAbsPower**, the fitted target is $\log |s|$, where s is the power; a predicted value z is decoded as $\hat{w} = \text{sgn}(s)K_0 e^z$, with the known weight K_0 of the parent form. The final checkpoint and the checkpoint minimizing validation loss are reported separately. Test errors play no part in checkpoint selection. Error bars are sample standard deviations over training seeds on fixed data partitions, not confidence intervals over alternative partitions or modular-form families. The tables below use validation-selected checkpoints. The alternate checkpoint tables are supplied as separate snippets. These new weight errors must not be substituted for earlier log-target errors without updating the metric definition and captions.

2.1 Positive integer powers, all 30 coefficients

Experiment	Training loss	Val. loss	Test MAPE (%)	Test RMSE
η	1.65×10^{-4}	0.00137	0.753 ± 0.194	6.83 ± 1.77
$N = 1$	7.34×10^{-5}	5.18×10^{-5}	0.622 ± 0.107	66.1 ± 16.1
$N = 2$	3.24×10^{-4}	3.28×10^{-4}	0.837 ± 0.155	40.3 ± 19.6
$N = 3$	2.68×10^{-4}	1.26×10^{-4}	0.998 ± 0.473	40.3 ± 9.28
$N = 5$	7.18×10^{-4}	0.00295	1.89 ± 1.04	53.4 ± 19.7
$N = 7$	2.30×10^{-4}	9.08×10^{-4}	0.996 ± 0.116	22.7 ± 9.79

Table 5: CHL and eta results for positive integer powers, using coefficient positions 0 through 29, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Dataset sizes are 999, 1999 rows; exact ranges are recorded with the data. Optimizer: RMSProp. This table replaces manuscript Table 2; its percentages are not errors of logarithmic targets.

2.2 Negative integer powers, all 30 coefficients

Experiment	Training loss	Val. loss	Test MAPE (%)	Test RMSE
$N = 1$	7.03×10^{-5}	6.91×10^{-5}	0.641 ± 0.0720	81.3 ± 10.3
$N = 2$	3.99×10^{-5}	4.02×10^{-5}	0.520 ± 0.103	36.5 ± 7.13
$N = 3$	6.55×10^{-5}	4.93×10^{-5}	0.522 ± 0.143	28.3 ± 4.41
$N = 5$	2.00×10^{-4}	8.69×10^{-5}	0.836 ± 0.155	25.2 ± 2.31
$N = 7$	8.05×10^{-4}	8.47×10^{-5}	0.816 ± 0.196	12.7 ± 4.96

Table 6: CHL results for negative integer powers, using coefficient positions 0 through 29, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Each dataset contains 999 rows. All rows use Net3. Optimizer: RMSProp. This table replaces manuscript Table 3; its percentages are not errors of logarithmic targets.

2.3 Positive rational powers, all 30 coefficients

Experiment	Optimizer	Training loss	Val. loss	Test MAPE (%)	Test RMSE
$N = 1$	RMSProp	4.73×10^{-5}	5.77×10^{-5}	0.528 ± 0.0692	48.4 ± 2.32
$N = 2$	RMSProp	1.94×10^{-4}	3.77×10^{-4}	0.813 ± 0.106	47.1 ± 8.41
$N = 3$	RMSProp	2.44×10^{-4}	4.74×10^{-4}	1.45 ± 0.297	53.7 ± 8.27
$N = 5$	ADAM	1.52×10^{-4}	8.81×10^{-4}	1.18 ± 0.603	15.2 ± 2.88
$N = 7$	SGD	8.96×10^{-5}	0.00645	2.18 ± 0.708	3.59 ± 0.420

Table 7: CHL results for the stored sample of positive rational powers, using coefficient positions 0 through 29, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Each dataset contains 2000 rows. Repeated rational powers are kept in the same data partition. All rows use Net3. The recorded optimizer choices are ADAM, RMSProp, SGD, as shown per row. This table replaces manuscript Table 4; its percentages are not errors of logarithmic targets.

2.4 Negative rational powers, all 30 coefficients

Experiment	Optimizer	Training loss	Val. loss	Test MAPE (%)	Test RMSE
$N = 1$	RMSProp	4.44×10^{-5}	3.91×10^{-5}	0.459 ± 0.0500	47.3 ± 6.84
$N = 2$	RMSProp	4.95×10^{-4}	2.97×10^{-5}	0.442 ± 0.0442	26.8 ± 4.86
$N = 3$	RMSProp	0.00373	1.19×10^{-4}	1.35 ± 0.0552	40.3 ± 7.52
$N = 5$	ADAM	1.38×10^{-5}	1.28×10^{-5}	0.461 ± 0.115	8.57 ± 1.45
$N = 7$	RMSProp	6.33×10^{-5}	4.84×10^{-5}	0.601 ± 0.142	13.4 ± 2.23

Table 8: CHL results for the stored sample of negative rational powers, using coefficient positions 0 through 29, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Each dataset contains 2000 rows. Repeated rational powers are kept in the same data partition. All rows use Net3. The recorded optimizer choices are ADAM, RMSProp, as shown per row. This table replaces manuscript Table 5; its percentages are not errors of logarithmic targets.

2.5 Positive integer powers, fixed 15-coefficient slice

Experiment	Training loss	Val. loss	Test MAPE (%)	Test RMSE
η	7.32×10^{-5}	5.61×10^{-4}	0.711 ± 0.170	4.95 ± 2.02
$N = 1$	1.06×10^{-4}	1.06×10^{-4}	0.697 ± 0.182	85.3 ± 21.9
$N = 2$	7.75×10^{-5}	4.30×10^{-4}	0.857 ± 0.0345	44.5 ± 4.11
$N = 3$	1.10×10^{-4}	9.66×10^{-5}	0.694 ± 0.205	39.0 ± 5.94
$N = 5$	2.03×10^{-4}	3.26×10^{-4}	0.912 ± 0.290	29.6 ± 6.98
$N = 7$	1.15×10^{-4}	2.92×10^{-4}	0.838 ± 0.235	16.4 ± 4.63

Table 9: CHL and eta results for positive integer powers, using the recorded fixed slice of 15 of the first 30 coefficient positions, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Dataset sizes are 999, 1999 rows; exact ranges are recorded with the data. All rows use Net3. Optimizer: RMSProp. This table replaces manuscript Table 6; its percentages are not errors of logarithmic targets.

2.6 Negative integer powers, fixed 15-coefficient slice

Experiment	Training loss	Val. loss	Test MAPE (%)	Test RMSE
$N = 1$	9.63×10^{-5}	7.92×10^{-5}	0.694 ± 0.186	78.4 ± 17.0
$N = 2$	4.72×10^{-5}	5.40×10^{-5}	0.493 ± 0.108	30.8 ± 5.85
$N = 3$	9.63×10^{-5}	6.74×10^{-5}	0.736 ± 0.277	36.1 ± 8.05
$N = 5$	1.92×10^{-4}	8.32×10^{-5}	0.752 ± 0.197	22.4 ± 2.37
$N = 7$	6.84×10^{-4}	9.31×10^{-5}	0.743 ± 0.212	15.0 ± 7.59

Table 10: CHL results for negative integer powers, using the recorded fixed slice of 15 of the first 30 coefficient positions, using the checkpoint selected by validation loss. Test MAPE is the mean absolute percentage error of the signed modular weight; test RMSE is in weight units. Predictions are transformed back from the fitted target before either test metric is evaluated. Losses are means in the fitted target space, as defined in the experimental protocol. Test entries show means across training seeds on a fixed split, with sample standard deviations when at least two seeds are available; the full study uses three seeds per row. Training budget: 10000 rounds. Each dataset contains 999 rows. All rows use Net3. Optimizer: RMSProp. This table replaces manuscript Table 7; its percentages are not errors of logarithmic targets.